

MANIKANTA NEERUKATTU (MANI N)

Senior Software Engineer | Applied AI Engineer | .NET, Azure & Agentic AI Systems

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PROFESSIONAL SUMMARY

Senior Software Engineer with 10 years of experience in the Microsoft ecosystem, progressing from full-stack enterprise development into Applied AI Engineering. Specializes in designing and shipping production-grade AI systems — Advanced RAG pipelines, multi-agent orchestration, and LLM-powered automation — that solve real business problems rather than stay as prototypes.

Strong backend foundation in .NET 8, ASP.NET Core, and Azure, now extended with hands-on depth in Azure AI Foundry, Semantic Kernel, and agentic frameworks. Has delivered AI-integrated systems across financial services, healthcare, and ERP, with consistent focus on reliability, scalability, and observable production behavior. Currently completing a Master of Science in AI/ML at Liverpool John Moores University, UK.

SKILLS

AI Engineering: Azure AI Foundry · Azure OpenAI (GPT-4, GPT-4o, Embeddings, Function Calling) · Semantic Kernel · Microsoft Agent Framework · LangChain · LangGraph · LlamaIndex · AutoGen · CrewAI · Ollama · HuggingFace · Mistral · RAG & Advanced RAG · Hybrid Search · Semantic Re-ranking · Prompt Engineering · LLM Evaluation (RAGAS) · MCP (Model Context Protocol) · Vector Databases (Azure AI Search, pgvector, ChromaDB) · Multi-Agent Orchestration · Agent Memory Architectures

Backend & System Design: C# · .NET 8/10 · ASP.NET Core · Python · FastAPI · Microservices Architecture · Event-Driven Systems · REST APIs · Azure SDK Integration · Domain-Driven Design · Clean Architecture

Cloud & Infrastructure: Microsoft Azure (Functions · API Management · Blob Storage · Service Bus · Application Insights · Key Vault) · Azure DevOps · GitHub Actions · Docker · OpenTelemetry · Distributed Tracing · Structured Logging

Data & Storage: PostgreSQL · SQL Server · Azure SQL · pgvector · Azure Data Factory · Databricks · PySpark

Frontend: Angular · React.js · Blazor · TypeScript · RxJS

Testing: xUnit · NUnit · Cypress · Playwright · SonarQube

PROFESSIONAL EXPERIENCE

Senior Software Engineer — AI & Applied Intelligence

Jun 2023 – Present

Insightsoftware India Pvt Ltd | Hyderabad, India

Led the AI transformation of enterprise financial disclosure and reporting systems. Responsible for taking AI features from design through to production, working across the full stack from Azure infrastructure to API surfaces to frontend integration.

- Designed and built an Advanced RAG pipeline using Azure AI Foundry, Azure OpenAI, and Azure AI Search — combining hybrid retrieval (keyword + vector), semantic re-ranking, intelligent chunking, and query expansion for high-accuracy compliance document processing.
- Engineered a multi-agent reporting system using Semantic Kernel, AutoGen, and LangGraph, with specialized agents for data retrieval (SQL/API), LLM reasoning, and structured report generation; reduced manual reporting effort by 60%.
- Built and integrated MCP (Model Context Protocol) enterprise connectors enabling AI agents to interact with SQL databases, REST APIs, and external SaaS systems — supporting natural language queries, dynamic function calling, and real-time schema exploration.
- Implemented memory-driven agentic workflows with vector memory and persistent context, enabling multi-turn document-aware AI interactions across long-horizon financial tasks.
- Developed a multi-model routing layer, directing different LLMs to specialized tasks (reasoning, extraction, summarization) to balance cost, latency, and output quality.
- Built scalable .NET 8 microservices integrating Azure OpenAI and AI Foundry APIs, with clean architecture, resilience patterns, and Azure API Management.
- Created prompt evaluation frameworks to systematically measure and improve LLM output quality, reducing hallucinations across model upgrades.

- Set up and maintained CI/CD pipelines (Azure DevOps) with automated quality gates including test coverage enforcement and static analysis.
- Instrumented services with OpenTelemetry, distributed tracing, and structured logging, providing operational visibility across distributed AI workflows.

Tech Stack: Azure AI Foundry · Azure OpenAI · Semantic Kernel · LangGraph · AutoGen · MCP · .NET 8 · Azure Functions · Azure AI Search · pgvector · React · Azure DevOps

Software Technologist

Apr 2022 – Jun 2023

Philips | Bangalore, India

Delivered enhancements to a regulated healthcare platform, integrating AI capabilities and strengthening API security and reliability.

- Integrated Azure Cognitive Services and Azure OpenAI to enable intelligent document processing and clinical decision support features within the platform.
- Built secure, high-throughput APIs using ASP.NET Core and Azure API Management, with OAuth2/OIDC authentication and input validation aligned to healthcare data compliance requirements.
- Developed real-time communication modules (chat and video) using SignalR, integrated into patient and clinician-facing web applications.
- Established automated regression testing pipelines using Cypress, cutting manual QA cycles and improving release stability.

Tech Stack: .NET Core · React · Angular · Azure Services · Azure OpenAI · SQL Server · Cypress

Senior Member Technical

Nov 2019 – Mar 2022

ADP Pvt Ltd | Hyderabad, India

Developed and maintained microservices components for ADP's enterprise payroll and HCM platform serving global clients.

- Built and owned microservices-based platform APIs using .NET Core and Azure, integrating multiple enterprise data sources and downstream systems.
- Designed event-driven workflows using Azure Service Bus and Azure Functions to automate payroll processing tasks, reducing manual operational overhead.
- Improved test coverage and CI/CD pipeline reliability, contributing to a more consistent delivery cadence across the team.

Tech Stack: .NET Core · Azure Functions · Azure Service Bus · Application Insights · SQL Server

Software Engineer

Oct 2016 – Nov 2019

Option Matrix Infotech | Hyderabad, India

- Built end-to-end ERP and CRM solutions using ASP.NET MVC, Angular, and SQL Server across multiple client engagements.
- Participated in Agile delivery cycles, authored unit tests, and contributed to code reviews.

KEY AI PROJECTS

1. Enterprise Advanced RAG Platform

A production-grade document intelligence platform built for legal, finance, and consulting use cases.

- Implemented hybrid retrieval (BM25 + dense vector search) with semantic re-ranking, significantly improving precision over pure vector search on long-form enterprise documents.
- Engineered chunking and embedding pipelines using sliding window and semantic chunking strategies tuned for financial and legal document structures.
- Built query rewriting and context enrichment stages to clarify and expand user queries before retrieval, improving answer groundedness and faithfulness.
- Deployed via Azure Functions with API Management; integrated a RAGAS evaluation harness for ongoing retrieval quality measurement.

Tech Stack: Azure AI Foundry · Azure OpenAI · Azure AI Search · .NET 8 · Semantic Kernel · pgvector

2. Agentic AI Reporting System

An autonomous multi-agent orchestration system for enterprise financial reporting.

- Designed a three-agent pipeline: Data Retrieval Agent (SQL/API via MCP), Analysis Agent (LLM reasoning), and Report Generation Agent (structured document output).
- Built an MCP tool interaction layer enabling agents to query enterprise databases using natural language and invoke APIs dynamically at runtime.
- Implemented with Semantic Kernel + AutoGen; currently migrating to Microsoft Agent Framework (MAF) for enterprise lifecycle management.

Tech Stack: Semantic Kernel · AutoGen · MCP · Azure OpenAI · .NET 8 · LangGraph

3. CareFlow AI-Powered Clinical Consultation System

End-to-end AI system automating clinical consultation workflows in a healthcare setting.

- Built an agentic clinical workflow engine handling patient intake, consultation documentation, and care plan generation using Azure OpenAI and Semantic Kernel.
- Implemented context-aware conversation management with persistent vector memory, maintaining clinical context across multi-turn patient-provider sessions.
- Developed a real-time alert and escalation layer via Azure Service Bus to route critical flags to appropriate care teams.

Tech Stack: .NET 8 · Azure OpenAI · Semantic Kernel · Azure Functions · Azure Service Bus · React

4. Financial Data Intelligence — XBRL/EDGAR Extraction & Anomaly Detection

A proof-of-concept financial intelligence platform targeting SEC EDGAR filings.

- Built an XBRL extraction pipeline to retrieve, parse, and normalize structured financial data from SEC EDGAR at scale (Azure Functions + Service Bus + PostgreSQL + Redis + Blob Storage).
- Designed anomaly detection APIs using numerical cross-period validation and statistical benchmarking against industry peer data.
- Produced a full HLD/LLD design covering ingestion architecture, orchestration, and evaluation strategy.

Tech Stack: Azure Functions · Service Bus · PostgreSQL · Redis · Blob Storage · Python · .NET 8

EDUCATION

Master of Science — Artificial Intelligence & Machine Learning

Expected 2026

Liverpool John Moores University, United Kingdom | Currently Pursuing

Postgraduate Diploma — Generative AI & Applied Machine Learning

2025

IIIT Bangalore | CGPA: 3.85 / 4.0

Bachelor of Technology — Electronics & Communication Engineering

2016

JNTU Kakinada, India

CERTIFICATIONS

- Microsoft Certified: Azure AI Engineer Associate (AI-102) In Progress, Expected 2026
- Executive PG in AI/ML — IIIT Bangalore, 2025

References available upon request